Adrien Carrou

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Education

Johns Hopkins University (JHU) M.S. in Electrical and Computer Engineering

San Jose State University (SJSU) B.S. in Computer Engineering

Work Experience

Staff Software Development Engineer - Intuitive Machines **Dec 2023 – Present**

- Develop and maintain C/C++ flight software for spacecraft mechanism control across a multi-processor LLP/HLP architecture, including stepper motor controllers for solar array drive and beam articulation mechanisms, sensor I/O, and command/telemetry interfaces.
- Diagnose cross-system flight software issues by tracing telemetry chains, autonomous fault-detection monitors, and stored command sequences across software bus boundaries.
- Write Google Test unit tests for motor control logic, service routines, and housekeeping telemetry; debug segmentation faults and integration failures.
- Verify ground-system command and telemetry database entries against C struct layouts at the bit level to support cross-subsystem command-table updates.
- Build Python-based hardware-in-the-loop and software-in-the-loop test infrastructure to exercise full mode-transition state machines and validate motor actuation end-to-end.
- Coordinate feature integration across a distributed team using Git submodule workflows, including merge-conflict resolution and multi-repo branch strategy.

Founder and Tech Lead - Nexus Analytica https://www.nexusanalytica.net/ **June 2024 – Present**

- Founded company specializing in multispectral and hyperspectral imaging systems.
- Lead embedded firmware and hardware development for sensor integration and acquisition pipelines.
- Designed multi-layer PCBs and FPGA-based acquisition logic for high-throughput sensing applications.
- Built custom firmware optimized for client-specific reliability and performance requirements.

Test Automation Engineer - Maxar Technologies **May 2022 – Dec 2023**

- Built simulation and test-automation pipelines for satellite flight software verification, including configuration management and reproducible-run procedures.
- Wrote Python tooling for log parsing, test harness setup, and result aggregation across simulated and real-hardware runs.
- Authored configuration, runbook, and integration documentation for downstream test engineers.

Leadership Experience and Projects

Club Leadership - SJSU Robotics and Cube3 **Sep 2021 – Aug 2023**

- Led multidisciplinary teams of 11-15 students across robotics and small-satellite projects; owned intelligent systems work including path tracking and positional accuracy, and guided subsystem integration through launch preparation.

Astraeus-Library https://astraeus-library.github.io/ **May 2023 – April 2025**

Open-source embedded avionics resources | C/C++, Embedded Systems, PCB Design, Firmware

- Developed firmware and multi-layer PCB design for Astraeus-I, an avionics development board built around an ARM Cortex-M class MCU with onboard sensor acquisition, data logging, and fault handling.
- Contributed sensor drivers to libhal, an open-source embedded hardware abstraction layer providing a unified API across hardware interfaces including GPIO, SPI, I2C, UART, and ADC for portable embedded C/C++ development.
- Authored documentation to make the library usable without lab access.

Skills

Programming Languages: C/C++, Python, Bash, Tcl, MATLAB, VHDL, Verilog

Frameworks & Platforms: cFE/cFS, FreeRTOS, Linux, Ubuntu, Rocky Linux, Docker, ROS, Google Test, CMake

Hardware & Tools: Multi-layer PCB design, Altium, KiCad, FPGA development, Xilinx, Intel FPGA, embedded MCUs, STM32, ESP32, nRF, Arduino, I2C, SPI, UART, CAN, BLE, Git, GitLab, submodules, merge request workflows